BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Merge Sort

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does insertion sort do with the next unsorted item?

2. What is ROM commonly used to store?

A) An unsaved document
B) The current mouse position
C) A running game's score
D) Startup instructions

3. "Cache is a remote Internet server that replaces CPU registers." Correct this. Explain why.

4. A pupil restarts a computer to open saved work. "RAM keeps every file forever." Correct this. Why is secondary storage needed?

Today's objectives

Explain the steps and required conditions for merge sort. Apply the algorithm or method to supplied data.

Key vocabulary - Read, cover, recall

Merge sort
Divide-and-conquer sort that splits and then merges sorted sublists.

Split
Divide a list into smaller parts.

Merge
Combine sorted sublists while preserving order.

Singleton
Sublist containing one element.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Divide and conquer	Merge sort repeatedly splits a list into smaller halves until each sublist has one item.
Single items are sorted	A one-element list is already in order by itself.
Merge in order	Take the smaller front item from two sorted sublists; copy it into the merged list.
Complete the sort	Repeat merging until there is one fully sorted list; do not confuse splitting with swapping.

I Do - One worked example

Show the major splitting and merging stages of merge sort for [8,3,5,1,7,2,6,4]. [6 marks]

1. Split into halves

2. [8,3,5,1] and [7,2,6,4], then split until single items remain.

3. Merge sorted pairs: [3,8], [1,5], [2,7], [4,6].

4. Merge quarters: [1,3,5,8], [2,4,6,7], then final [1,2,3,4,5,6,7,8]. [6]

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Split [6,3,8,1] into singletons and merge into ascending order. [5]

2. Merge two sorted sublists [2,7,9] and [1,3,10]. [4]

3. Explain why each half must be sorted before it can be merged in the usual merge step. [3]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Select which first element should be taken from two sorted lists.

```
array left = [4,10]
array right = [2,8]
if ... then
print(left[0])
else
print(right[0])
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Split [7,2,5,1] to singletons and write the two sorted pairs. [4]

Q3. Merge the already sorted lists [1,4,9] and [2,3,8]. State the resulting list. [3]

Q4. Explain what divide-and-conquer means in merge sort. [3]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Binary search can safely discard half only when data is sorted. ☐ T ☐ F
2. Merge sort repeatedly splits a list into smaller halves until each sublist has one item. ☐ T ☐ F
3. A one-element list is already in order by itself. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Place the next item in the correct position in the sorted part. Shift larger items right. ☐

Q2 D - Startup instructions ☐

Q3 Cache is in or near the CPU. It gives faster access to often-used data and instructions than main memory. ☐

Q4 Secondary storage keeps files and programs without power. Save the work there before restarting. ☐

Code First - Model

```
array left = [4,10]
array right = [2,8]
if left[0] <= right[0] then
print(left[0])
else
print(right[0])
endif
```

Output: 2

We Do - Check the model

1. [6,3] and [8,1] [1]; [6],[3],[8],[1] [1]; [3,6],[1,8] [1]; [1,3,6,8] [2]. ☐

2. Choose 1,2,3,7,9,10 [3]; resulting [1,2,3,7,9,10] [1]. ☐

3. The merge chooses the smallest front item [1]; this is safe only when each sublist is ordered [1], producing a sorted merged list [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Split [7,2] and [5,1] [1]; singletons [7],[2],[5],[1] [1]; sorted pairs [2,7] [1] and [1,5] [1]. ☐

Q3 [1,2,3,4,8,9] [3; 1 for first three, 1 for last three, 1 for completely correct]. ☐

Q4 Divide the list into smaller subproblems [1], solve them by sorting/merging sublists [1], combine to obtain one ordered result [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: